

Quantum Simulation

From Hamiltonians to Adiabatic Computation

Outline of Topics

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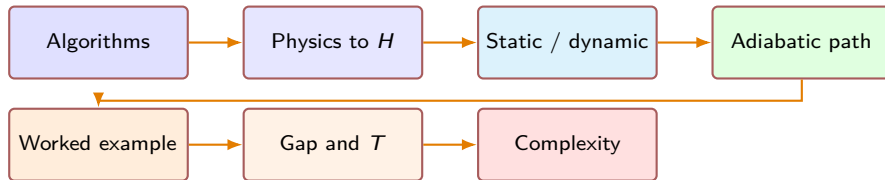
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- Quantum simulation connects a physical model to a computational task
- Shared language: algorithms, Hamiltonians, and implementation methods
- This talk focuses on adiabatic Hamiltonian computation and its bottlenecks

Content outline

Physical intuition first; then problem classes; then adiabatic methods, gap, runtime, and complexity.

Roadmap



Content outline

Adiabatic paths are developed in detail. Digital primitives (Trotter, LCU, QSP) appear only as later tools.

What Is an Algorithm?

- Finite procedure: input $\rightarrow$ instructions $\rightarrow$ output
- Independent of a particular machine realization
- “Quantum” changes the execution model, not this definition

Content outline

Operational definition; contrast with a device-specific program.

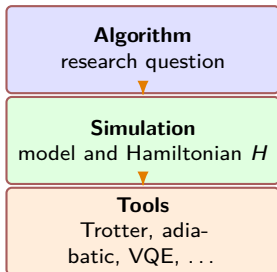
Quantum Algorithms at a Glance

- ① State preparation on qubits
- ② Unitary evolution: gate circuits or Hamiltonian control
- ③ Measurement and extraction of observables

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Three-part skeleton only; no gate-set or circuit-depth discussion yet.

Three Layers



Simulation is not a single gate.
It is a model-centered path:
question $\rightarrow H \rightarrow$ method.

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Layer 1 asks; layer 2 represents the physics; layer 3 implements $U(t)$, $H(s)$, or $f(H)$.

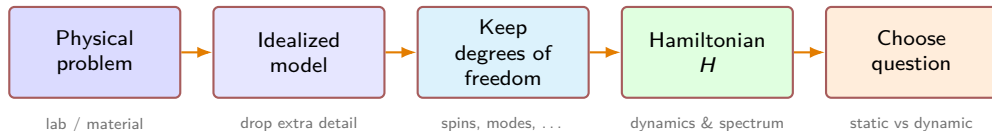
What Quantum Simulation Means

- Use a quantum system to study another quantum model
- Digital, analog, or hybrid realizations
- Common starting point: fix H and the retained degrees of freedom

Content outline

Typical models: Hubbard, molecules, spin lattices, circuit modes.

From Physical Problem to Mathematical Model



Content outline

A model is a deliberate compression: what is kept versus discarded.

Representative Physical Problems

Physical setting	Typical model	Degrees of freedom
Magnets / spin glasses	Ising / Heisenberg	spins on a lattice
Correlated electrons	Hubbard	electrons + spin
Molecules	electronic structure	active orbitals
Cavities / circuits	oscillator / LC	flux and charge

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The same material can map to different models, depending on the question.

Classical Hamiltonians

- Generalized coordinates and momenta (q, p)
- Legendre transform from the Lagrangian to H
- For usual conserved systems, H is the total energy

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}$$

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H is both the energy function and the generator of classical motion.

Two Roles of the Classical H

- ① Energy on phase space
- ② Generator of classical time evolution

Important distinction

Minimizing energy (equilibria / optimization) is not the same as following dynamics.

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Minimization maps to combinatorial Ising costs; dynamics maps to trajectories.

Quantum Hamiltonians

- Canonical quantization: $[q, p] = i\hbar$
- Dynamics: $i\hbar \partial_t |\psi\rangle = \hat{H} |\psi\rangle$
- Spectrum: $\hat{H} |E_n\rangle = E_n |E_n\rangle$

Content outline

H both sets the allowed energies and generates the evolution of the state.

Why H Is Central in Simulation

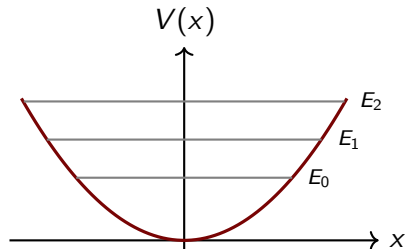
- Almost every task is a function of one Hamiltonian:
 - time evolution e^{-iHt}
 - ground state and spectrum
 - scheduled path $H(s)$
 - more general $f(H)$
- Choosing Trotter or VQE before fixing H is premature

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Fix the model and H first; then choose the question and the method.

Harmonic Oscillator

- $H = \omega(a^\dagger a + \frac{1}{2})$
- Equally spaced levels
- Creation / annihilation operators
- Language of cavities, phonons, circuit modes

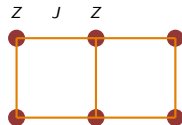


Content outline

Physical intuition for bosonic modes; no full derivation here.

Spins and the Ising Model

- Two-level degrees of freedom on a lattice
- ZZ couplings and local fields
- Ferro- / antiferromagnetism
- Natural map to qubits and optimization

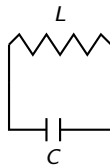


Content outline

Algorithmic details of Ising return in the adiabatic worked example.

LC Circuits: Classical to Quantum

- Energy: $H = Q^2/(2C) + \Phi^2/(2L)$
- Quantize with $[\hat{\Phi}, \hat{Q}] = i\hbar$
- Recovers a harmonic oscillator
- Frequency $\omega = 1/\sqrt{LC}$



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Circuit energy $\rightarrow$ canonical pair $(\Phi, Q) \rightarrow$ quantum spectrum.

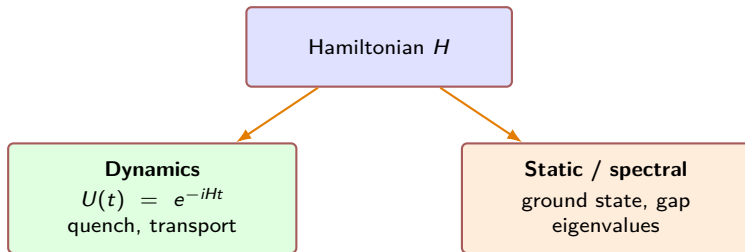
Anharmonicity and Qubit Encoding

- A harmonic spectrum is equally spaced, so the lowest two levels are not an isolated qubit
- Josephson nonlinearity distorts the level spacings
- Logical chain: circuit $\rightarrow$ classical $H \rightarrow$ quantization $\rightarrow$ anharmonicity if a qubit is required

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Engineering details of superconducting qubits are left for a later, hardware-focused discussion.

Two Questions from One Hamiltonian



Content outline

The same H supports two research paths. Response functions often sit in between.

Problem Classes in Quantum Simulation

- Dynamic: post-quench response, transport, observable evolution
- Static: ground energies, optimal Ising configurations
- Intermediate: low-lying spectrum, gaps, state preparation

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Classify the question before choosing a method.

Two Main Approaches

- ① **Time evolution:** build or approximate e^{-iHt} or $f(H)$
- ② **Adiabatic:** slow path $H_B \rightarrow H_P$, stay in the instantaneous ground state

Focus ahead

Adiabatic computation, worked example, Qiskit outline, then gap, runtime, and complexity.

Content outline

Trotter, LCU, and QSP are named only as tools for the evolution approach.

Adiabatic Idea in One Slide

- ① Start from an easy H_B with a known ground state
- ② Interpolate slowly to the problem Hamiltonian H_P
- ③ Sufficient gap and time $\Rightarrow$ ground state of H_P

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Encode the problem in a Hamiltonian path rather than enumerating configurations.

Problem Scope for Adiabatic Methods

- ① Combinatorial optimization mapped to classical Ising
- ② Ground-state preparation of physical models
- ③ Hardware that natively implements a schedule $H(s)$ (annealing / analog)

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Short examples: MaxCut, transverse-field Ising, analog annealers.

Why Use Adiabatic Computation?

- Transparent physical picture of the computation
- Natural match to analog field schedules
- Digitally expressible via Trotterized paths (digital adiabatic)

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Less natural when the target is a precise e^{-iHt} or a general $f(H)$.

The Path $H(s)$

- Standard linear interpolation

$$H(s) = (1 - s) H_B + s H_P, \quad 0 \leq s \leq 1$$

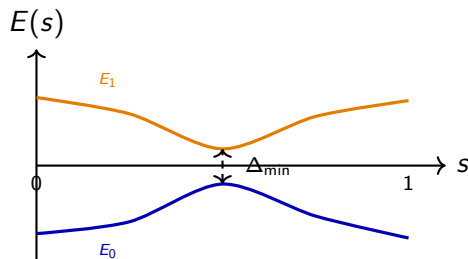
- Example schedule: $s(t) = t/T$
- s is not physical time; it mixes the two Hamiltonians

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$H(s)$ is linear in s ; the eigenvalues generally are not.

Instantaneous Gap and Runtime T

- $\Delta(s) = E_1(s) - E_0(s)$
- $\Delta_{\min} = \min_s \Delta(s)$ is the bottleneck
- Heuristic scale: $T \propto 1/\Delta_{\min}^2$



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Avoided crossing: the two lowest levels approach, then separate, without crossing.

Two-Qubit Setup

- Make H_B , H_P , the gap, and T concrete
- System: two spin-1/2 particles (two qubits)
- Parameters: $h_x = 1$, $J = 1$, $h_z = 1/4$

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The example is small enough that the full spectrum can be seen numerically.

Driver Hamiltonian H_B

- $H_B = -h_x(X_1 + X_2)$
- Easy ground state $|++\rangle$, prepared by Hadamards on $|00\rangle$
- Role: a known, preparable starting point

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Transverse field; product eigenstates of X .

Problem Hamiltonian H_P

- $H_P = -JZ_1Z_2 - h_z(Z_1 + Z_2)$
- Target ground state $|00\rangle$ for $J, h_z > 0$
- Ferromagnetic coupling plus a longitudinal field that lifts the degeneracy

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Optimization reading: the ground configuration is the solution.

Numerical Path Result

- Gap minimum near $s_* \simeq 0.62$
- $\Delta_{\min} \simeq 0.66$
- Endpoint gaps alone do not locate the bottleneck

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Scan the instantaneous spectrum along the whole path $H(s)$.

From Gap to a Runtime Estimate

- $\|H_P - H_B\|_2 \simeq 2.37$
- Heuristic with $\varepsilon \sim 10^{-2}$: $T_{\text{heur}} \sim 5 \times 10^2$
- For $n = 2$, $T \sim 40$ already yields near-unit success probability

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Distinguish a heuristic bound from a practical runtime checked by dynamics.

Role of Qiskit

- Shared implementation stack for digital circuits
- Turns the adiabatic path into a reproducible workflow
- Environment: `conda activate qcomputing`

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Close the loop from the Hamiltonian path to executable steps.

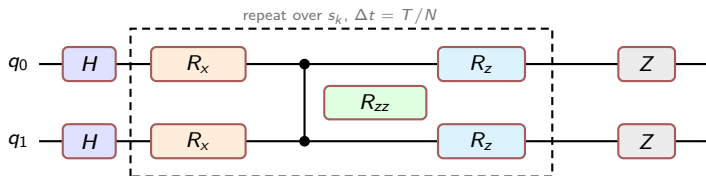
Code Steps (1): Hamiltonians and Gap

- 1 Build H_B and H_P with `SparsePauliOp`
- 2 Form $H(s) = (1 - s)H_B + sH_P$
- 3 Scan s and extract $\Delta_{\min}$ and s_*

Content outline

Skeleton only; scripts in `snippets/adi_ising_*.py`.

Code Steps (2): Circuit and Readout



Content outline

Hadamard preparation, first-order Trotter layers, then Z readout.

Sample Execution Outcome

- With $T = 40$ and $N = 200$: $P(|00\rangle) \approx 0.9999$
- Aer shots concentrate on the bitstring 00
- The same run produces the path-gap plot

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Success probability is a check of the digital adiabatic schedule.

Gap Is Not a Single Number

- Full gap versus the gap inside the reachable symmetry sector
- Avoided crossing versus a true level crossing
- Symmetry-breaking noise can remove protection

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The smallest spectral distance alone is not a sufficient criterion.

Three Meanings of Required Time

- ① Heuristic / order-of-magnitude estimate
- ② Sufficient bound under explicit assumptions
- ③ Practical time from dynamics and a success threshold

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These three numbers usually disagree. Local scheduling: slow down near the bottleneck.

Scaling with System Size

- Polynomial gap closing: $\Delta_{\min}(n) \sim n^{-\alpha}$
- Exponential closing: $\Delta_{\min}(n) \sim e^{-cn}$
- With $T \sim 1/\Delta_{\min}^2$, the runtime may become exponential

Content outline

Qubit count alone is not a measure of difficulty.

Milder vs. Hard Regimes

Milder

- Gapped paths
- Away from criticality
- Some solvable models

Hard

- Near critical points
- Tunneling / exponentially small gaps
- Generic local ground-energy problems

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In the worst case, estimating the ground energy of a local Hamiltonian is QMA-complete.

Three Complexity Layers

- ① Hardness of the **ground-state problem**
- ② Hardness of a **particular adiabatic path**
- ③ Cost of a **digital implementation** of that path

Message

Keep these three layers separate when assessing feasibility.

Key Takeaways

- ① Simulation starts by fixing H and the degrees of freedom
- ② Static and dynamic questions are distinct for the same H
- ③ Adiabatic computation is a path $H(s)$; the gap sets the time
- ④ Complexity follows $\Delta_{\min}(n)$ and the problem class, not n alone

- Adaptive schedules that slow down near a small gap
- Noise and symmetry breaking along an analog or digital path
- Comparison with variational routes such as QAOA and VQE

Content outline

Related notes: `QS&QA_Foundation/QS&QA_Foundation.tex`

Questions